Recommendation Pipeline - Design

- Dataset: 100,000 public MovieLens interactions from the ashraq/movielens_ratings mirror, plus its movie catalog.

- The mirror has no authoritative event timestamps; timestamps are deterministically generated (NumPy seed 42).

- Interactions retained: 100,000; quarantined: 0.

- Chronological holdout: 80,000 older training interactions, 20,000 newer test interactions.

- Stages: ingest, validate/quarantine, chronological preparation, feature snapshots, two models, and ranking evaluation.

- Features are stored as Parquet entity snapshots; the interaction matrix is stored as sparse NPZ.

- Models: interaction-count popularity baseline and personalized truncated-SVD collaborative filtering.

- GitHub repository: https://github.com/QuickLearnerQ/DM4ML

Recommendation Pipeline - Results

model	RMSE	MAE	Precision@10	Recall@10	NDCG@10	Coverage	Diversity
popularity	1.0134	0.7617	0.0273	0.227	0.1181	0.0018	0.8183
truncated_svd	1.0262	0.7848	0.0038	0.0279	0.0148	0.026	0.7631

- Ranking metrics use K=10 and held-out ratings of at least four stars.

- Ranking uses a deterministic sample of up to 1,000 relevant users.

- Coverage is the recommended share of the item catalog; diversity is pairwise genre Jaccard distance.

- The full comparison table is available in outputs/model_comparison.csv.

Recommendation Pipeline - Limitations

- A time-based split better approximates future interactions than a random split.

- Invalid rows are retained in data/errors with rejection reasons.

- JSONL tracking stores model parameters, metrics, and artifact paths without a tracking server.

- The Python scheduler can execute this workflow once or at a configured interval; an optional daily Airflow DAG runs the same workflow.

- MovieLens cold-start users/items, popularity bias, and offline metrics limit generalization.

- Mirror timestamps are synthetic when the source mirror does not provide event times; temporal evaluation does not represent real event chronology for this input.